
Joint Biomedical Named Entity Recognition and Relation Extraction on ADKG and MDKG

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Abstract

1 Extracting structured knowledge from biomedical text is a foundational task in
2 clinical NLP. We apply SpERT, a span-based transformer model for joint named
3 entity recognition (NER) and relation extraction, to two biomedical knowledge
4 graph datasets: the Alzheimer’s Disease Knowledge Graph (ADKG) and the Mental
5 Disorder Knowledge Graph (MDKG). Using SciBERT as the backbone, SpERT
6 achieves NER micro F1 scores of 70.87% and 81.24% and relation extraction micro
7 F1 scores of 40.21% and 49.63% (with named entity classification) on ADKG
8 and MDKG respectively. We conduct exploratory data analysis to characterize
9 entity-relation distributions, and examine edge cases including near-miss entities,
10 confused relation types, and the effect of long-range dependencies on model
11 performance. As an extension, we investigate transfer learning by training on the
12 combined datasets, finding that while aggregate performance declines modestly,
13 certain shared relation types such as *characteristic_of* benefit substantially from
14 the additional training data, suggesting that selective data mixing may be a more
15 effective strategy than naive concatenation.

1 Introduction

17 Biomedical knowledge graphs (KGs) encode relationships between clinical concepts such as diseases,
18 drugs, genes, and symptoms. Creating these graphs from free text requires solving NER and relation
19 extraction jointly since the two tasks are tightly connected: relation extraction assumes correct entity
20 detection, and entity classification benefits from relational context.

21 Span-based models offer a natural framework for this joint task. SpERT Eberts and Ulges (2020)
22 enumerates candidate entity spans, classifies them, and then scores all pairs of predicted entities
23 for relation membership in a single forward pass. Paired with SciBERT Beltagy et al. (2019), a
24 language model pretrained on scientific literature, SpERT is well-suited to biomedical text where
25 domain vocabulary and complex multi-word expressions are common.

26 We apply SpERT to ADKG, a dataset focused on Alzheimer’s disease, and MDKG, a broader medical
27 KG focusing on mental disorders. Our goals are to (1) characterize each dataset through exploratory
28 data analysis; (2) evaluate SpERT’s performance using standard precision, recall, and F1 metrics; (3)
29 investigate sources of performance gaps such as overlapping entities and long-range dependencies;
30 and (4) assess whether combining both datasets as auxiliary data improves performance via transfer
31 learning.

2 Methods

2.1 Datasets

ADKG and MDKG are biomedical sentence-level datasets annotated with entity spans and directed relations. Each sentence contains character-level entity offsets and entity-ID-based relation references. Table 1 summarizes the dataset statistics. A detailed description of both datasets can be found [here](#).

Table 1: Summarized dataset statistics. MDKG exhibits higher entity and relation density than ADKG.

Split	ADKG			MDKG		
	Train	Dev	Test	Train	Dev	Test
Sentences	5,605	1,206	1,220	4,825	941	912
Entities	20,859 total			28,660 total		
Relations	5,496 total			10,560 total		
Ents/sent	2.60			4.29		
Rels/sent	0.68			1.58		
Entity types	6			9		
Relation types	8			9		

Data conversion. SpERT requires token-level entity offsets and integer-indexed relation references. We wrote a conversion script that tokenizes each sentence by whitespace, maps character-level spans to token indices, and replaces entity ID references with list indices. Zero entities or relations were skipped across all splits of both datasets.

2.2 Model

SpERT performs joint NER and relation extraction in a single forward pass. Given a sentence tokenized by BERT, it enumerates all spans up to a maximum length, classifies each as an entity or non-entity, and scores all entity pairs for relation membership using a filtered set of candidate pairs. We use `allenai/scibert_scivocab_cased` as the backbone, which is pretrained on biomedical and scientific literature and is well-matched with our datasets.

2.3 Experimental Setup

Single-dataset training. Both single-dataset models were trained for 20 epochs using predefined parameters (e.g., AdamW optimizer with learning rate 5×10^{-5} , warmup ratio 0.1, batch size 16, and maximum span size of 10 tokens). Negative entity and relation counts were set to 100 each. Models were evaluated on the held-out test set using micro-averaged precision, recall, and F1, both with and without named entity classification (NEC), where NEC additionally requires correct entity types for a relation to count as correct. Training was performed on an NVIDIA RTX 5070 Ti (16 GB VRAM) under Windows Subsystem for Linux (WSL2).

Combined training. As an extension, we combine the ADKG and MDKG training sets (10,430 sentences total) to investigate whether cross-domain biomedical data improves performance. A unified types file is constructed as the union of both type sets, resulting 11 entity types and 10 relation types. The combined model uses identical hyperparameters and is evaluated separately on each test set, allowing direct comparison against the single-dataset baselines.

3 Exploratory Data Analysis

Figure 1 shows that *disease* is the dominant entity type in both datasets. ADKG includes a vague *other* category (1,540 instances) that encompasses entities that do not fit cleanly into more specific types, making it inherently harder to model. MDKG introduces nine distinct entity types including *Health_factors*, *physiology*, *region*, and *signs*, providing richer semantic annotation at the cost of increased class complexity.

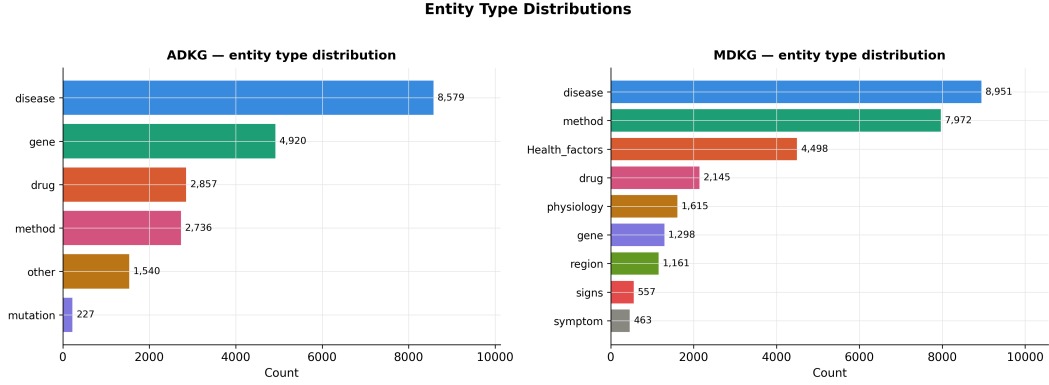


Figure 1: Entity type distributions for ADKG (left) and MDKG (right). Disease is the dominant type in both datasets.

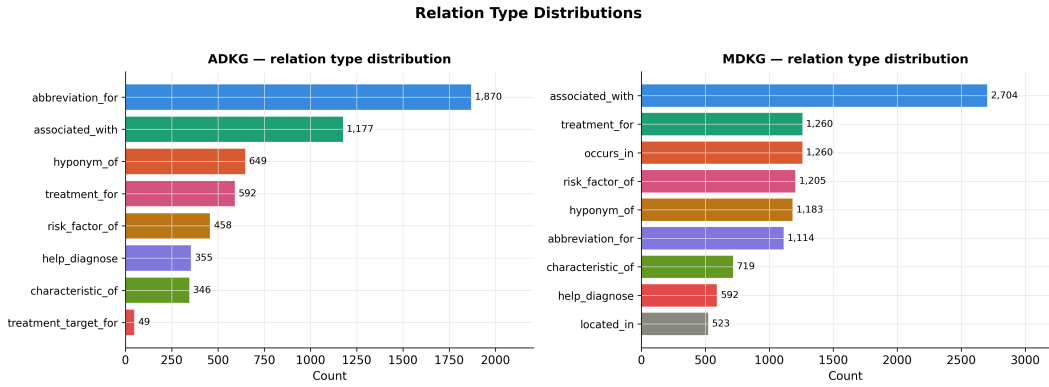


Figure 2: Relation type distributions for ADKG (left) and MDKG (right). ADKG is heavily skewed toward *abbreviation_for* while MDKG has a more balanced distribution.

Figure 2 shows a strong class imbalance in ADKG, where *abbreviation_for* accounts for 1,870 of 5,496 relations (34%). MDKG is more balanced, with *associated_with* as the most common relation type (2,704 instances, 25.6%). Figure 3 shows that same-type entity pairs dominate both datasets (*disease*→*disease* is the most common pair in both), which can encourage the model to rely on entity type rather than contextual cues when predicting relations.

4 Results

4.1 NER Performance

As shown in Table 2, SpERT achieves a NER micro F1 of 70.87% on ADKG and 81.24% on MDKG. The performance difference is largely attributable to ADKG’s *other* category (F1 30.35%), which is semantically ambiguous. *Disease* is the best-performing type in both datasets, as expected given its high frequency.

4.2 Relation Extraction Performance

Table 3 shows relation extraction results with NEC. *Abbreviation_for* is the strongest relation in both datasets (ADKG: 69.86%, MDKG: 76.68%) due to its predictable pattern (e.g., parenthetical abbreviations). *Associated_with* is weak in both datasets (ADKG: 11.41%, MDKG: 36.45%), as it is broad and frequently confused with more specific relation types.

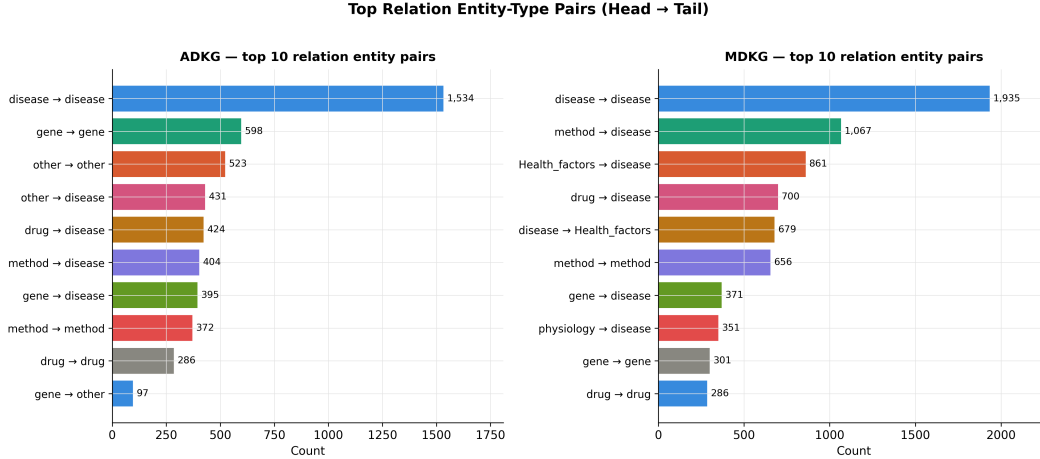


Figure 3: Top 10 head→tail entity type pairs for ADKG (left) and MDKG (right). Same-type pairs, especially disease→disease, dominate in both datasets.

Table 2: NER performance by entity type on ADKG and MDKG test sets, sorted by ADKG F1. *Disease* is the strongest type in both datasets.

Entity Type	ADKG			MDKG		
	P	R	F1	P	R	F1
disease	81.66	85.22	83.40	89.87	92.51	91.17
mutation	73.47	81.82	77.42	—	—	—
gene	72.25	71.07	71.66	78.64	75.70	77.14
drug	69.13	64.64	66.81	72.63	80.24	76.25
method	53.02	56.16	54.55	80.12	81.43	80.77
other	33.33	27.86	30.35	—	—	—
Health_factors	—	—	—	75.93	78.96	77.41
region	—	—	—	73.20	81.75	77.24
physiology	—	—	—	68.35	66.67	67.50
signs	—	—	—	65.52	63.33	64.41
symptom	—	—	—	37.78	33.33	35.42
Micro	70.82	70.91	70.87	80.23	82.27	81.24

4.3 Edge Cases and Performance Gaps

Overlapping entities. Overlapping entity pairs are rare in both datasets (ADKG: 0.36%, MDKG: 0.28%), indicating that SpERT’s non-overlapping constraint is not a significant source of error.

Long-range dependencies. Approximately 22.8% of ADKG relations and 28.6% of MDKG relations span more than 10 tokens between head and tail entities (Table 4). Long-range relations are harder for BERT-based models to capture, and likely contribute to MDKG’s lower relation F1 (49.63%) relative to its NER F1 (81.24%).

Near-miss entities. A common failure is predicting the correct entity type but with a span off by one token. For example:

Text: "Importantly, the antidepressant **fluoxetine** restores both adult neurogenesis ..."
Type: [drug]
Gold: [fluoxetine] (tokens 3–4)
Pred: [antidepressant fluoxetine] (tokens 2–4)

Text: "**Endogenous Abeta peptide** promote Abeta oligomerization tendency of spiked ..."
Type: [gene]

Table 3: Relation extraction performance by relation type with NEC on ADKG and MDKG test sets, sorted by ADKG F1. *Abbreviation_for* is the strongest relation in both datasets.

Relation Type	ADKG			MDKG		
	P	R	F1	P	R	F1
abbreviation_for	64.15	76.69	69.86	73.17	80.54	76.68
help_diagnose	33.33	21.15	25.88	48.15	54.93	51.32
hyponym_of	22.93	38.71	28.80	52.79	66.67	58.92
treatment_for	25.51	31.65	28.25	49.03	52.05	50.50
characteristic_of	34.78	16.67	22.54	32.39	21.70	25.99
risk_factor_of	24.14	23.73	23.93	56.93	49.06	52.70
associated_with	14.29	9.49	11.41	38.83	34.35	36.45
treatment_target_for	0.00	0.00	0.00	—	—	—
occurs_in	—	—	—	54.04	61.27	57.43
located_in	—	—	—	47.69	40.79	43.97
Micro	39.47	40.97	40.21	50.04	49.23	49.63

Table 4: Long-range dependency statistics.

	ADKG	MDKG
Avg token distance	7.1	8.9
Max token distance	48	140
Relations >10 tokens	1,254	3,019
% long-range	22.8%	28.6%

97 **Gold:** [Endogenous Abeta peptide] (tokens 0–3)

98 **Pred:** [Abeta peptide] (tokens 1–3)

99 **Confused relation types.** The most frequent confusion in both datasets is between *associated_with*
100 and more specific types such as *risk_factor_of* or *characteristic_of*. For example:

101 **Text:** "Overall, social disconnectedness predicted higher subsequent perceived isolation,
102 which in turn predicted higher depression symptoms ..."

103 **Relation:** [perceived isolation -> depression symptoms]

104 **Gold:** [associated_with]

105 **Pred:** [risk_factor_of]

106 4.4 Transfer Learning Results

107 Table 5 shows that combining both datasets leads to modest performance degradation overall, with
108 NER F1 dropping 3.03 points on ADKG and 1.77 points on MDKG, and relation F1 (with NEC)
109 declining by 0.65 and 0.48 points respectively. Despite the overall decline, type-level inspection
110 reveals meaningful variation: relation types with consistent cross-dataset semantics such as *hy-*
111 *ponym_of* (+11.39 on ADKG) and *help_diagnose* (+9.24 on MDKG) benefit substantially from
112 combined training, while NER degrades more uniformly — particularly for dataset-specific types
113 such as *signs* (−12.56) and *region* (−5.76) on the ADKG test set, which have zero support in ADKG
114 and effectively add noise to the output distribution. The inconsistency between NER and relation
115 extraction degradation suggests that entity type boundaries are more dataset-specific, while some
116 relation semantics transfer more readily across biomedical subdomains.

117 5 Discussion

118 SpERT with SciBERT achieves reasonable performance on both biomedical KG datasets, particularly
119 for NER. The performance gap between ADKG and MDKG could be driven by three main factors:
120 (1) ADKG’s noisy *other* entity category, which lacks a consistent definition; (2) ADKG’s class
121 imbalance in relation types, with *treatment_target_for* having only 49 training examples; and (3)
122 MDKG’s higher entity density (4.29 vs. 2.60 entities/sentence) providing more training signal per
123 sentence.

Table 5: Detailed F1 comparison between single-dataset and combined model for NER (top) and relation extraction with NEC (bottom), sorted by single-dataset ADKG F1. Δ denotes change from single-dataset baseline; positive values indicate improvement.

	ADKG F1			MDKG F1		
	Single	Comb.	Δ	Single	Comb.	Δ
<i>Named Entity Recognition</i>						
disease	83.40	83.25	−0.15	91.17	91.69	+0.52
mutation	77.42	70.33	−7.09	—	—	—
gene	71.66	70.07	−1.59	77.14	72.30	−4.84
drug	66.81	67.18	+0.37	76.25	74.02	−2.23
method	54.55	49.21	−5.34	80.77	79.61	−1.16
other	30.35	30.05	−0.30	—	—	—
health_factors	—	—	—	77.41	74.56	−2.85
region	—	—	—	77.24	71.48	−5.76
physiology	—	—	—	67.50	61.67	−5.83
signs	—	—	—	64.41	51.85	−12.56
symptom	—	—	—	35.42	31.25	−4.17
Micro	70.87	67.84	−3.03	81.24	79.47	−1.77
<i>Relation Extraction (with NEC)</i>						
abbreviation_for	69.86	71.07	+1.21	76.68	74.45	−2.23
help_diagnose	25.88	18.35	−7.53	51.32	60.56	+9.24
hyponym_of	28.80	40.19	+11.39	58.92	55.35	−3.57
treatment_for	28.25	28.57	+0.32	50.50	47.74	−2.76
characteristic_of	22.54	27.85	+5.31	25.99	30.30	+4.31
risk_factor_of	23.93	18.84	−5.09	52.70	52.92	+0.22
associated_with	11.41	9.29	−2.12	36.45	33.93	−2.52
treatment_target_for	0.00	0.00	—	—	—	—
occurs_in	—	—	—	57.43	59.41	+1.98
located_in	—	—	—	43.97	45.52	+1.55
Micro	40.21	39.56	−0.65	49.63	49.15	−0.48

Relation extraction remains the harder subtask in both datasets, with F1 approximately 30 points lower than NER. The gap between NEC and non-NEC relation F1 indicates that entity type errors propagate into relation predictions, motivating further work on improving entity classification for rare types. Long-range dependencies are a meaningful source of error, particularly in MDKG where 28.6% of relations span more than 10 tokens. Future work could explore cross-sentence context to better capture distant entity relationships. The near-miss pattern suggests that span boundary detection could be improved through better sub-word tokenization.

The combined training experiment reveals that naive dataset concatenation is insufficient for effective transfer between biomedical subdomains. While aggregate micro F1 declines for both datasets, the type-level analysis tells a more nuanced story: relation types with consistent cross-dataset semantics (*hyponym_of*, *characteristic_of*) benefit from additional training data, while entity type performance degrades due to type mismatch between datasets. These findings suggest that selective data mixing — for example, upsampling shared relation types or excluding dataset-specific entity types from the combined training objective — may be a more effective transfer strategy than simple data concatenation. Future work could train a single shared model with separate output layers for each dataset, allowing the model to share general biomedical knowledge without forcing it to reconcile conflicting type definitions.

References

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- Eberts, M. and Ulges, A. (2020). Span-based joint entity and relation extraction with transformer pre-training. In *ECAI 2020*.